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Lecture 09: Dynamic OT Part II

Last Time

- Static vs Dynamic OT comparison.
  - Static (Kantorovich): Defined by couplings $\alpha \rightarrow \beta$; Transport is one-shot, static; Find coupling minimizing distance; OT as min-cost (soft) matching between distributions.
  - Dynamic: Considers flows ρ_t, v_t transporting $\alpha 1 \beta$; Transport is time-dependent, dynamic; Find flow minimizing energy; OT as shortest-path (geodesic) in probability space.
- Dynamic OT Formulation: Benamou-Brenier:

$$W_2(\alpha, \beta)^2 = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \, \rho_t(x) \, dx \, dt,$$

  where ρ_t, vt are measure and velocity field, and are subject to:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0 \text{ (continuity equation)}; \ \rho_0 = \alpha, \ \rho_1 = \beta \text{ (boundary conditions)}$$

- Benamou Brenier Intuition: $\rho_t(t)$: measure at time t and $v_t(x)$: time-varying vector field; $\{v_t, \rho_t\}_{t=0}^1$ defines a path in $\mathcal{P}(\mathcal{X})$; path length: $\int$ of velocity through time; W_2: length of shortest path.
- Displacement vs Mixture Interpolation:
  - Displacement Interpolation: $\rho_t^{\text{disp}} = [(1-t)\text{Id} + tT]_{\#}\alpha$; Interpolates locations of points.
  - Mixture Interpolation: $\rho_t^{\text{mix}} = (1-t)\alpha + t\beta$; Interpolates masses of points.

-> formalize intuition with benamou brenier formulation. with this form the constraint is a pde. think of the dynamic of the problem to learn.
compare displacement interp with the naive mixture interp

Today Today we will continue working on dynamic OT with 2 directions.

- Gradient Flows:
  - Warm-up: to motivate, will start with the Euclidean case, with Euclidean gradient flows.
  - From Euclidean to Wasserstein flows.
  - The JKO scheme (one way to time discretize a gradient flow).
- Schrödinger Bridges:
  - Dynamic OT over path space (alternative view as an optimization over distributions over path space).
  - The Schrödinger problem.
  - Connections to entropic OT, Benamou–Brenier (and see how everything fits together).
- Applications in ML.

1 Part I: Gradient Flows

Why gradient flow? Why do we need gradient flows?

- Benamou-Brenier formulation:

$$W_2(\alpha,\beta)^2 = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \, \rho_t(x)\, dx\, dt,$$

  where ρ_t, vt are measure and velocity field, and are subject to:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0 \text{ (continuity equation)}; \ \rho_0 = \alpha, \ \rho_1 = \beta \text{ (boundary conditions)},$$

  which consists in finding the entire path (i.e., finding the entire trajectory at once).

- But what if finding the entire path $\{\rho_t, v_t\}$ is too costly? In fact, finding the entire path can sometimes be infeasible or impractical, and maybe the problem could be solved gradually instead of doing all at once.
- What if we don't explicitly know β? (But we know what we would like β to satisfy.) An analogous situation is gradient descent (GD), where we initially don't know what the desired weights are (which would be β) but we know what they have to satisfy.

Gradient Flows Intuition

- Instead of finding entire path at once, solve the problem gradually (by evolving over time, instead of finding all at once as in Benamou-Brenier).
- Instead of target β, we have a functional $F(\rho)$ we want to minimize.
- Analogy: gradient descent, we know where we start and objective, and want to move towards objective.
- Use cases:
  - Diffusion models — distribution evolves by diffusion and learned drift.
  - Generative sampling — gradual transformation of noise into data.
  - Regularization flows — denoising, inpainting, PDE-based priors.
  - Physics-inspired learning — population dynamics, mean-field games.

Warmup: Gradient Flows in $\mathbb{R}^d$

- Gradient descent:

$$x_{n+1} = x_n - \gamma \nabla f(x_n), \qquad \gamma > 0 \ \text{ (stepsize)},$$

  where the stepsize controls how fast we move between iterations, and x_n are vectors in some $\mathbb{R}^d$.

- Taking the limit $\gamma \to 0$ leads to an ordinary differential equation called the *gradient flow*:

$$\dot{x}(t) = -\nabla f(x(t)).$$

- This is a curve $x(t)$ in $\mathbb{R}^d$ that starts at $x(0) = x_0$ and moves by choosing, at each instant, the direction of steepest descent of f. As we can see in the visualization, the gradient flow is the smooth limit of gradient descent when stepsize$\to$ 0, i.e., it is the continuous version of gradient descent.
- Pros:
  - Simplified analysis (no choice of step-size, etc).
  - Can often prove convergence of gradient flow in cases where proving things about gradient descent is hard.
  - Connects to ODE/SDE theory.
- But:
  - Cannot directly run the flow in continuous time! Practical algorithm requires time discretization.

Discretizing Gradient Flows

- There's two main ways to discretize the gradient flow $\dot{x}(t) = -\nabla f(x(t))$:
  - Explicit: relying on direct computation of ∇f. E.g., Euler's method:

    $$x_{t+1} = x_t - \tau \nabla f(x_t) \text{ (Forward Euler Method).}$$

  - Implicit: defining steps as solution to an optimization problem. E.g.,

    $$x_{t+1} = \arg\min_x \left\{ f(x) + \tfrac{1}{2\tau}\|x - x_t\|^2 \right\} \text{ (Backward/Implicit Euler Method).}$$

- Both share the same principle: make local steps that balance:
  i. Reducing f.
  ii. Staying close to current location x_t.

  (Spoiler: will use same principle for distributional grad flows.)
- But implicit methods are much easier to generalize (e.g. to non-Euclidean settings, ∇f can be not available/expensive, etc.) and is more stable in practice.

From $\mathbb{R}^d$ to $\mathcal{P}(\mathbb{R}^d)$

- What if instead of a single point x_t, we evolve a probability distribution ρ_k?
- Let $F : \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}^d$ be a functional over distributions, and ρ_0 a starting point.
- Can we define a gradient flow of F? I.e., a curve in $\mathcal{P}(\mathbb{R}^d)$ starting at ρ_0 and evolving in the direction of maximal descent of F?
- We need: notion of gradient (for the explicit method) or notion of distance (for the implicit method).
- Much easier to start from distance. (For the gradient, it requires taking derivatives of functionals with respect to measures, which is not straightforward).
- Recall implicit Euler scheme:

  $$x_{t+1} = \arg\min_x \left\{ f(x) + \tfrac{1}{2\tau}\|x - x_t\|^2 \right\}.$$

- Limit as $\tau \to 0$: continuous gradient flow in $\mathbb{R}^d$.
- Let's do the same in $\mathcal{P}(\mathbb{R}^d)$. First step: choose a meaningful distance.
- Good choice of distance: Wasserstein distance W_2.
- Leads to the JKO scheme: a time-discretized version of the Wasserstein gradient flow in $\mathcal{P}(\mathbb{R}^d)$.

JKO scheme

- One implicit Euler step in Wasserstein space:

  $$\rho_{t+1} = \arg\min_\rho \{\mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho_t)\},$$

  where $\mathcal{F}(\rho)$ is the function we want to minimize, $W_2^2(\rho, \rho_t)$ is the squared distance from the current distribution ρ_t, and $\tau > 0$ is the time-step parameter.
- This approach ensures a balance between:
  - Lowering the energy $\mathcal{F}(\rho)$.

  - Staying close to the previous iterate (in Wasserstein distance).

  (Same as Euclidean gradient flows.)

- For small τ: W_2 weighs more, ensures staying close to current ρ_t, smaller step size.
- As $\tau \to 0$, the discrete scheme $\to$ continuous gradient flow.

Note that we won't cover here how the arg min is solved computationally (in practice, different methods can be used to solve it, some using the convexity of the problem). How to turn the arg min problem into computationally efficient method will be covered in the discussion sessions.

Historical Context & Importance of JKO

- Jordan–Kinderlehrer–Otto (1998):
  - Introduced the variational formulation of diffusion as a gradient flow in Wasserstein space.
  - Showed that the heat and Fokker-Planck equations can be recovered as the gradient flows of certain energies.
- Impact:
  - Sparked development of "Otto calculus": treating probability measures as an (∞-dim) Riemannian manifold.
  - Provided a powerful bridge between PDE theory, optimal transport, and variational methods.
- Practical importance:
  - Unified framework for many PDEs (heat, Fokker–Planck).
  - Basis for many modern numerical schemes (proximal / variational time-stepping).
  - Influential in ML and generative modeling: diffusion models, regularized flows, entropy minimization.

Gradient Flows: Big Picture

- Many PDEs = gradient flows of energies $\mathcal{E}(\rho)$.
- In Wasserstein space $((\mathcal{P}(\mathbb{R}^d), W_2)$, the JKO scheme makes this precise.
- Key idea: PDE evolution = iterated minimization of energy + transport cost.

Back to continuous time

- JKO Scheme Wasserstein space: $\rho^{k+1} \in \arg\min_\rho \left\{\mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2\left(\rho, \rho^k\right)\right\}$.
- What happens as $\tau \to 0$?
- We need a notion of derivative of $\mathcal{F}(\rho)$ to study this.
- For a functional $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$, the *first variation* of $\mathcal{F}$ at ρ, denoted $\frac{\delta \mathcal{F}}{\delta \rho}$, satisfies

$$\frac{d}{d\varepsilon} \mathcal{F}(\rho + \varepsilon\chi)\Big|_{\varepsilon=0} = \int \frac{\delta \mathcal{F}}{\delta \rho}(x)\, d\chi, \quad \forall \chi \in \mathcal{P}(\mathcal{X}).$$

- Intuition: $\frac{\delta \mathcal{F}}{\delta \rho} \approx$how much $\mathcal{F}$ changes if we add an infinitesimal amount of mass at location x.

- One can show that as $\tau \to 0$, the solution of

$$\rho^{k+1} \in \arg\min_{\rho} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2\Big(\rho, \rho^k\Big) \right\}$$

  corresponds to the solution of:

$$\partial_t \rho_t \;=\; \nabla \cdot \left(\rho_t \, \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right).$$

- We can briefly explain the meaning of each term:
  - $\partial_t \rho_t$ is the time derivative of the density—i.e., the distribution changes over time.
  - $\frac{\delta \mathcal{F}}{\delta \rho}$ is the first variation of the functional—"potential"/ driving force field.
  - $\nabla \frac{\delta \mathcal{F}}{\delta \rho}$ is the gradient of that potential—i.e., a local velocity vector that points to the direction of maximal change of $\mathcal{F}$.
  - $\rho_t \, \nabla \frac{\delta \mathcal{F}}{\delta \rho}$ is the mass flux—how much mass moves through each point.
  - $\nabla \cdot$ converts flux into net inflow/outflow.
- We can define the Wasserstein Gradient as

$$\nabla_W \;\triangleq\; \nabla \cdot \left(\rho \, \nabla \frac{\partial(\cdot)}{\partial \rho} \right),$$

  then the gradient flow can be written as $\partial_t \rho_t \;=\; \nabla_W \mathcal{F}(\rho_t)$.

In other words, we have seen that in the limit as $\tau \to 0$, the discrete JKO scheme converges to a continuous-time evolution. In particular, the density ρ_t is then governed by a PDE. This PDE for $\partial_t \rho_t$ can be interpreted as a continuity equation associated with a velocity field in probability space. A key takeaway is that structures from Euclidean space have analogues in the space of probability measures.

Back to Continuous Time: Dynamic Formulation of OT (Benamou–Brenier)

Setup. Let $\mathcal{X}$ be the domain. We consider a time-dependent *measure* ρ_t and a *vector field* v_t.

Benamou–Brenier dynamic formulation.

$$W_2^2(\alpha, \beta) \;=\; \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \|v_t(x)\|^2 \, \rho_t(x) \, dx \, dt \quad \text{subject to} \quad \begin{cases} \partial_t \rho_t + \nabla \cdot (\rho_t \, v_t) = 0, \\ \rho_0 = \alpha, \quad \rho_1 = \beta. \end{cases} \tag{1}$$

Wasserstein Gradient Flow

Gradient flow equation.

$$\partial_t \rho_t \;=\; \nabla \cdot \left(\rho_t \, \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right). \tag{2}$$

Notes.

- This is a PDE on ρ.
- Does this look familiar? It is the continuity equation for the vector field

$$v_t(x) = -\nabla \left(\frac{\delta \mathcal{F}}{\delta \rho}(x) \right).$$

Takeaway.

- JKO scheme $\implies$ continuous gradient flow.
- The "direction of steepest descent" in Wasserstein space is encoded by the first variation of $\mathcal{F}$.

Examples of Energies & PDEs

Key power of Wasserstein gradient flows: they recover/unify many well-known PDEs.

Choice of energy functional $\mathcal{F}$ and resulting PDE $\big(\partial_t \rho_t = \nabla_W \mathcal{F}(\rho_t)\big)$:

$$\textbf{Heat Equation:} \qquad \mathcal{F}(\rho) = \int_{\mathcal{X}} \rho \log \rho \, dx, \qquad \partial_t \rho = \Delta \rho. \tag{3}$$

$$\textbf{Porous Medium:} \qquad \mathcal{F}(\rho) = \frac{1}{m-1} \int_{\mathcal{X}} \rho^m \, dx \qquad (m > 1), \qquad \partial_t \rho = \Delta(\rho^m). \tag{4}$$

$$\textbf{Fokker–Planck:} \qquad \mathcal{F}(\rho) = \int_{\mathcal{X}} \rho \log \rho \, dx \; + \; \int_{\mathcal{X}} V(x)\, \rho(x) \, dx, \qquad \partial_t \rho = \nabla \cdot \big(\rho \, \nabla(\log \rho + V)\big). \tag{5}$$

2 Three Canonical Energies

The below are particular instances of three very nice functions of gradient flows, they have well-behaved f-function, and there are three classes such that they are internal energy, potential energy, and interaction energy.

Internal Energy.

$$\mathcal{F}(\rho) \;=\; \int_{\mathbb{R}^d} f(\rho(x)) \; \mathrm{d}x, \qquad \frac{\delta \mathcal{F}}{\delta \rho}(x) \;=\; f'\big(\rho(x)\big).$$

Intuition: penalizes local density (entropy, porous medium, etc. In interaction energy, we can define a general class of PDEs, and we can pick whatever we want and create a combined, internal, potential and interaction to get a total energy additive model).

Potential Energy.

$$\mathcal{V}(\rho) \;=\; \int_{\mathbb{R}^d} V(x)\, \rho(x) \; \mathrm{d}x, \qquad \frac{\delta \mathcal{V}}{\delta \rho}(x) \;=\; V(x).$$

Intuition: external field / drift (confinement, data term). We can catch loss functions with potential energy.

Interaction Energy.

$$\mathcal{W}(\rho) \;=\; \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} W(x-y)\, \rho(x)\, \rho(y) \; \mathrm{d}x \; \mathrm{d}y, \qquad \frac{\delta \mathcal{W}}{\delta \rho}(x) \;=\; (W * \rho)(x).$$

Intuition: nonlocal attraction/repulsion. We can catch repulsion with the interaction energy.

Total Energy (additive model).

$$\mathcal{E}(\rho) \;=\; \underbrace{\mathcal{F}(\rho)}_{\text{internal}} \;+\; \underbrace{\mathcal{V}(\rho)}_{\text{potential}} \;+\; \underbrace{\mathcal{W}(\rho)}_{\text{interaction}}.$$

Resulting Wasserstein Gradient Flow (PDE).

$$\partial_t \rho \;=\; \nabla \cdot \Big(\rho \, \nabla \Big[f'(\rho) + V + (W * \rho) \Big] \Big).$$

With the above three families of functions, we have access to model any functions we would like, being able to model many different PDEs from them. A way to think about this is that they are the most comprehensive origins of PDEs in their own right.

3 Part II: Schrödinger Bridges

3.1 Motivation and Intuition

- Dynamic OT gives sharp displacement but can be unstable in noisy data.This is a generalization that comes from particle physics theory, it comes from the intuition that dynamic OT can help define paths, but it can be unstable and be noisy.
- Idea: allow diffusive noise in the paths. if we move a particle and if it was noisy or moving in an imperfect way, we cannot transpose the geodesics too much.
- Schrödinger bridge (SB): "Most likely stochastic process (close to Brownian motion) that matches observed marginals." The Schrodinger Bridge is essentially the problem of finding the most stochastic processes. It still matches the marginal however.
- ML connections:
  - Diffusion Schrödinger Bridge (De Bortoli et al., 2021, etc.). This paper has taken Schrodinger Bridges and applied them to diffusion models in the visual space.
  - Score-based diffusion models $\sim$ SBs with learned guidance.

3.2 Dynamic Formulation on Path Space

We are taking the route here that is closest to OT, where we solve the problem, just over a different space.

Consider the space of paths $\bar{\mathcal{X}} = C([0,1], \mathcal{X})$ and path measures $\bar{\pi} \in \mathcal{P}(\bar{\mathcal{X}})$. We will consider paths between points and look at all possible paths over our points. It also includes other maybe longer paths, and basically all paths. Let $P_t : \bar{\mathcal{X}} \to \mathcal{X}$ be the evaluation map $P_t(\gamma) = \gamma(t)$. Given endpoint distributions $\alpha = \rho_0$ and $\beta = \rho_1$, define

$$\Pi(\alpha, \beta) \;=\; \left\{\bar{\pi} \in \mathcal{P}(\bar{\mathcal{X}}) : (P_0)_{\#}\bar{\pi} = \rho_0,\ (P_1)_{\#}\bar{\pi} = \rho_1\right\}.$$

Since we are doing something distributional , we will define a distribution over path space, and we will define the path over certain distributions. The dynamic OT over path space can be written as

$$W_2^2(\rho_0, \rho_1) \;=\; \min_{\bar{\pi} \in \Pi(\rho_0, \rho_1)} \int_{\bar{\mathcal{X}}} \mathcal{L}(\gamma)\ \mathrm{d}\bar{\pi}(\gamma), \qquad \mathcal{L}(\gamma) \;=\; \int_0^1 \|\dot{\gamma}(s)\|^2\ \mathrm{d}s.$$

Relation to static Kantorovich. Now we need to decide what trajectory we are flowing to, so we need to be more general. This is a whole trajectory where we take a distribution over path. We first look at the initial snapshot and see if they match the first snapshot over marginals.

If $\pi^\star$ is a static optimal plan, the optimal path measure $\bar{\pi}^\star$ gives mass only to geodesics joining pairs (x, y) in proportions prescribed by $\pi^\star$. For discrete measures,

$$\pi^\star \;=\; \sum_{i,j} \pi_{ij}\, \delta_{(x_i, y_j)}, \qquad \bar{\pi}^\star \;=\; \sum_{i,j} \pi_{ij}\, \delta_{\gamma_{x_i, y_j}},$$

where γ_{x_i, y_j} is the geodesic between x_i and y_j.

Intermediate marginals. The time-t snapshot measures $\rho_t := (P_t)_{\#}\bar{\pi}^\star$ coincide with the Benamou–Brenier solution. When we take snapshots and define measure, we must match Benamou-Brenier.

3.3 Schrödinger Problem

- Replace geodesics by noisy paths (Brownian motion). Here what we will use is Brownian motion in stead of geodesics.

- Regularize the path distribution by enforcing closeness to a diffuse reference path measure. We regularize geodesics in this scenario, technically our reference path measure is actually Brownian motion. We are minimizing the divergence.
- Reference path measure $\mathcal{K}$: the law of reversible Brownian motion (formally, the symmetric Wiener measure).

SB optimization.

$$\min_{\bar{\pi} \in \Pi(\rho_0, \rho_1)} \mathrm{KL}\big(\bar{\pi} \,\|\, \mathcal{K}\big).$$

This problem has a unique solution $\bar{\pi}_\varepsilon^\star$. It can be viewed as a mixture of Brownian bridges $\bar{\gamma}_{x,y}^\varepsilon$ (a "fuzzy geodesic") connecting x to y. We can think of it as a set of Brownian bridges, which is basically a crackled fuzzy geodesic graph visually.

Connections. Whenever you have some entropy or are doing some smoothing, we can imagine regularizing to some unique solution, such as how does this solution relate to some other parts of OT. As the noise level $\varepsilon \to 0$, Schrödinger bridges $\to$ OT on paths and $\bar{\gamma}_{x,y}^\varepsilon \to \delta_{\gamma_{x,y}}$. The SB solution is related to the *entropic OT* solution $\pi_\varepsilon^\star$ via Brownian bridges. For discrete measures:

$$\pi_\varepsilon^\star \;=\; \sum_{i,j} \pi_{\varepsilon,ij}^\star \, \delta_{(x_i, y_j)}, \qquad \bar{\pi}_\varepsilon^\star \;=\; \sum_{i,j} \bar{\pi}_{\varepsilon,ij}^\star \, \bar{\gamma}_{x_i, y_j}.$$

Entropic displacement interpolation. The intermediate marginal at time t is

$$\rho_t^\varepsilon \;=\; (P_t)_\# \bar{\pi}_\varepsilon^\star,$$

i.e., an *entropic* displacement interpolation, which can be seen (formally) as a Gaussian blurring of the usual displacement interpolation. Some intuition on the Schrodinger problem, is that one asks themselves what happens if they were to take the epsilon over noise. We incorporate epsilon, as it encompasses the Brownian motion.

Regularized Benamou–Brenier view. One can also arrive at the above interpolation by directly regularizing the Benamou–Brenier functional:

$$\min_{(\rho_t, v_t)_t} \int_0^1 \int_{\mathbb{R}^d} \Big(\|v_t(x)\|^2 + \tfrac{\varepsilon}{4} \|\nabla \log \rho_t(x)\|^2 \Big) \rho_t(x) \; \mathrm{d}x \; \mathrm{d}t, \quad \text{subject to the continuity equation.}$$

Intuition:

- $\|v_t(x)\|^2$ encodes kinetic energy (particle speed at x). As noise level goes down, the fizzy geodesics become actual geodesics.
- The diffusion penalty is high if ρ_t is sharply peaked; low if it is flat. Same intuition as other regularizations.
- This balances transport cost with diffusion: mass moves along velocity fields but may also spread like heat.
- Interpolates between sharp OT geodesics and noisy Schrödinger bridges. You can show that the solution, which is the Schrodinger Bridge, is just the entropic OT solution.

When Schrodinger Bridges are referenced in OT literature, an allusion can be made that we solve entropic OT using the Schrodinger Bridges. An example can be that when you are doing displacement interpolation, you can recover the Schrodinger Bridge between two bridges.

Side note: there is also a heat interpretation of the Schrodinger Problem.

4 ML Applications

4.1 Diffusion Schrödinger Bridge

Connections to diffusion models and score-based generative modeling; SB with learned guidance (e.g., De Bortoli et al., 2021). There are many papers with applications. One such paper is this one. It approaches the topic of applying Schrodinger Bridges to applications concerning visual modeling by solving multiple instances of Sinkhorn.

5 Gradient Flows for Dataset Optimization

5.1 Motivation: Dataset 'Shaping'

Another example is where we could have an initial dataset, and what we want is something that looks like the reference. A way to think is to think about gradient descent is to think of it as a way to optimize. In papers regarding this, many problems are framed as gradient flows. Goals for an initial dataset may include:

- Maximize similarity to another dataset (e.g., transfer learning). We want to use OT to maximize similarity to another dataset.
- Increase class separation.
- Hide protected/sensitive attributes.

5.2 One Framework

Let $\rho \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ be a joint distribution. We solve

$$\min_{\rho} \; F(\rho) \quad \Longrightarrow \quad \rho^\star \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}).$$

(See A.-M. & Fusi, *Dataset Dynamics via Gradient Flows in Probability Space*, ICML 2021, `https://arxiv.org/abs/2010.12760`.)

5.3 Gradient Flows on Probability Space

We seek to optimize over the (infinite-dimensional, non-Euclidean) space $\mathcal{P}(\mathcal{X} \times \mathcal{Y})$. The Wasserstein gradient flow for F is

$$\partial_t \rho(t) \;=\; -\nabla \cdot \left(\rho(t)\, \nabla \frac{\delta F}{\delta \rho}\big(\rho(t)\big) \right), \qquad \rho(0) = \rho_0.$$

A practical scheme uses time discretization and a finite-particle approximation:

$$z_{t+1}^{(i)} \;\leftarrow\; z_t^{(i)} \;-\; \lambda\, \nabla_{z^{(i)}} F(\hat{\rho}_t), \qquad i = 1, \ldots, n,$$

where each "particle" is a data point (e.g., an image).

5.4 Simple Dataset Dynamics

Objective: $F(\rho) = \mathrm{OTDD}(\rho, \beta)$ with a reference dataset β. Variants: Feature-driven, jointly-driven (fixed labels or variable labels), with optimizers such as SGD/Adam.

5.5 Application: Dataset 'Shaping'

Datasets need to be nearly separable, and when we have gradient flow, we just minimize distance and get our original gradient flow back. Given initial dataset ρ_0 and target (reference) $\rho^\star$, we may choose

$$F(\rho) \in \Big\{ \mathrm{OTDD}(\rho, \rho^\star), \;\; \mathcal{V}(\rho), \;\; \mathrm{OTDD}(\rho, \rho^\star) + \mathcal{V}(\rho) \Big\}$$

to achieve a target distribution that is close to $\rho^\star$ but also, e.g., linearly separable.

5.6 Application: Data-Driven Transfer Learning

Another example is we have an initial dataset, we start from the initial dataset and we flow it to a new dataset on what we want it to look like, and then train our new model based on the ending position of our initial dataset.
Usual (model-based) adaptation: train on source, fine-tune on target.
Here: *data-based* adaptation — flow source data toward target, then train on the flowed data.

Example evaluation settings:

- Datasets: MNIST, USPS, KMNIST, FashionMNIST.
- Setting: small target dataset (100 / class), larger source dataset (1000 / class).
- Ablations: target-only; target+source; target+mapped source; target+last-step flow; target+full flow trajectory; target+full flow trajectory + mapped.

References